

Pre-Layer: Template Reconstruction

Document Owner: Zernitsky Itamar **Status:** Design Reference **Scope:** Full technical specification, design rationale, accuracy discussion, and illustrative worked examples for the Pre-Layer template reconstruction phase.

How to read this document. Every conceptual section is followed by an illustrative example drawn from SITE-001 real data topology (13 weeks, 4 rooms, 18 providers). The example dataset is fully consistent throughout — the same providers, the same rooms, the same anomalies appear in every section so every formula connects back to the same ground truth.

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1. Why Template Reconstruction Exists

Hospitals do not schedule blocks from scratch each week. They operate on a **rotating template** — a repeating cycle of OR assignments that gets unrolled into a calendar. A 2-week rotation means Week 1 and Week 3 and Week 5 are all instances of Phase 0, while Week 2 and Week 4 and Week 6 are all instances of Phase 1.

This structure is never stored explicitly in the EHR. The EHR records what actually happened — which provider held which block in which room on which day — but not the underlying template. The template lives in the heads of the OR scheduling team.

The Pre-Layer's job is to **recover that template from the historical record**, without being told what it is. It must answer three questions:

Question 1 — How long is the cycle? Is it a 1-week, 2-week, 4-week, or longer rotation? This is the period detection problem.

Question 2 — What is the template? Given the cycle length, what is the canonical assignment for each (room, day, time) slot in each phase of the cycle?

Question 3 — Where did the template deviate? On which specific dates did the actual schedule differ from the canonical template? These deviations carry signal for downstream demand forecasting.

Everything downstream depends on getting these three answers right. If the detected period is wrong, every phase-aligned feature in Layer 1 will be misaligned. If the template is wrong, the optimizer warm-start will be wrong. If exceptions are missed, the `exception_rate` feature will misrepresent which slots are contested.

2. The Raw Data: EHR Block History Matrix

2.1 Input Records

The Pre-Layer receives one record per block per day across the historical range (default: 13 weeks).

Field	Type	Notes
block_historical_id	string	EHR's own identifier — may be inconsistent across weeks
site	string	Facility identifier
location_id	string	Room identifier
occurrence.start	datetime	Block start timestamp
occurrence.end	datetime	Block end timestamp
current_blockholder	string	provider_id, "Open", or "Closed"
service_line	string	Clinical service line of the room
manual_early_release	datetime	If provider released block early

2.2 Building the History Matrix

The flat list of records is transformed into a structured matrix:

- **Rows:** unique (site, room, day_of_week, time_slot) combinations — called **cells**
- **Columns:** weeks $0, 1, 2, \dots, H - 1$ where H is the number of historical weeks
- **Values:** provider_id (or "Open" / "Closed") for each cell × week combination

This is the **cell-week history matrix** X where $X[c][w]$ is the actual blockholder in cell c during week w .

SITE-001 · CELL-WEEK HISTORY MATRIX · H=13 weeks · 28 cells				
CELL DEFINITIONS (room · day · canonical start)				
C01	OR-101	Monday	07:00	Cardiac room
C02	OR-101	Tuesday	07:00	Cardiac room
C03	OR-101	Wednesday	07:00	Cardiac room
C04	OR-101	Thursday	07:00	Cardiac room
C05	OR-101	Friday	07:00	Cardiac room
C06	OR-102	Monday	07:00	Ortho room
C07	OR-102	Tuesday	07:00	Ortho room
C08	OR-102	Wednesday	07:00	Ortho room ← alternating cell
C15	OR-104	Monday	07:00	General (open pool)
...	(20 more cells)			

HISTORY MATRIX $X[\text{cell}][\text{week}]$								
Cell	W00	W01	W02	W03	W04	W05	...	W12
C01	Garcia	Garcia	Garcia	Garcia	Garcia	Garcia	...	Garcia
C02	Chen	Chen	Chen	Chen	Chen	Chen	...	Chen
C03	Garcia	Garcia	Garcia	Garcia	Garcia	Garcia	...	Garcia
C04	Garcia	[Chen]	Garcia	Garcia	Garcia	Garcia	...	Garcia
		↑ W01 exception: Chen covered Garcia's Thursday						
C05	Garcia	Garcia	Garcia	Garcia	Garcia	[Wright]...		Garcia
						↑ W05 exception		
C06	Thompson	Thompson	Thompson	Thompson	Thompson	Thompson	...	Thompson
C07	Kim	Kim	Kim	Kim	Kim	Kim	...	Kim
C08	Thompson	Patel	Thompson	Patel	Thompson	Patel	...	Thompson
	↑ ————— KEY CELL: clean 2-week alternation drives $T^*=2$ ————— ↑							
C15	Open	Open	Open	Open	Open	Open	...	Open
SUMMARY								
Total cells	: 28							
Stable cells	: 24 (same provider every week)							
Alternating cells	: 2 (2-week rotation pattern visible)							
Exception cells	: 3 (mostly stable, occasional deviation)							
Always-open cells	: 4							

2.3 The Alphabet

Each cell's values come from a finite alphabet $\mathcal{V}$ containing all possible blockholders: every provider_id, plus "Open" and "Closed". The alphabet size $V = |\mathcal{V}|$ matters for BIC scoring. For SITE-001: $V = 18 \text{ providers} + 2 = 20$.

V is the **global** alphabet size. Most cells have only 2–4 possible values in practice. The global V slightly overstates complexity cost but ensures consistent comparison across all cells and all candidate periods.

3. Period Detection via BIC

3.1 The Intuition

We want to find the rotation period T^* that best explains the history matrix X .

"Best explains" means: if we assume the schedule repeats every T weeks, how

often does the actual data match what the template would predict?

We test every integer period $T \in [1, \min(13, H)]$ and score each candidate using a Bayesian Information Criterion (BIC) that trades off fit quality against model complexity.

The fundamental trade-off: a longer period T can fit the data better (fewer deviations) but costs more to describe (the template has more entries). BIC penalizes complexity so we do not overfit to noise.

3.2 Phase Grouping

For a given candidate period T , partition the H historical weeks into T groups based on rotation phase:

$$W_{T,\phi} = \{ w \in 0, \dots, H-1 \mid w \bmod T = \phi \}$$

PHASE GROUPING · SITE-001 · H=13 weeks · Testing T=1, T=2, T=4
Under T=1 (no rotation – every week is in the same phase)
Phase 0 = {W00, W01, W02, W03, W04, W05, W06, W07, W08, W09, W10, W11, W12} → 13 observations
Under T=2 (2-week rotation)
Phase 0 = {W00, W02, W04, W06, W08, W10, W12} → 7 observations
Phase 1 = {W01, W03, W05, W07, W09, W11} → 6 observations
Under T=4 (4-week rotation)
Phase 0 = {W00, W04, W08, W12} → 4 observations
Phase 1 = {W01, W05, W09} → 3 observations
Phase 2 = {W02, W06, W10} → 3 observations
Phase 3 = {W03, W07, W11} → 3 observations
KEY DIAGNOSTIC – Cell C08 (OR-102 Wednesday) under each T:
Actual history:
W00:Thompson W01:Patel W02:Thompson W03:Patel W04:Thompson
W05:Patel W06:Thompson W07:Patel W08:Thompson W09:Patel
W10:Thompson W11:Patel W12:Thompson
Under T=1:

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|| All 13 weeks in one group
|| Modal = Thompson (7 times) ← Patel (6 times) = 6 deviations
||
|| Under T=2:
|| Phase 0 = W00,W02,W04,W06,W08,W10,W12 → all Thompson → modal=Thompson
|| Phase 1 = W01,W03,W05,W07,W09,W11 → all Patel → modal=Patel
|| Deviations = 0 ✓ Perfect fit
||
|| Under T=4:
|| Phase 0 = W00,W04,W08,W12 → all Thompson → 0 deviations
|| Phase 1 = W01,W05,W09 → all Patel → 0 deviations
|| Phase 2 = W02,W06,W10 → all Thompson → 0 deviations
|| Phase 3 = W03,W07,W11 → all Patel → 0 deviations
|| Also 0 deviations – but template cost is 2× higher than T=2
||
|| → T=2 explains C08 perfectly at half the template cost of T=4

```

3.3 Modal Template Value

For each cell c and phase ϕ under period T , the template assignment is the most frequently observed blockholder:

$$m_{c,\phi}^{(T)} = \arg \max_{v \in \mathcal{V}} |\{w \in W_{T,\phi} : X[c][w] = v\}|$$

Ties are broken by the most recently observed value.

3.4 Deviation Count

A deviation is any week where the actual blockholder differs from the modal value:

$$d_{c,\phi}^{(T)} = |W_{T,\phi}| - \max_{v \in \mathcal{V}} |\{w \in W_{T,\phi} : X[c][w] = v\}|$$

Total deviations across all cells and phases:

$$D(T) = \sum_{c \in \mathcal{C}} \sum_{\phi=0}^{T-1} d_{c,\phi}^{(T)}$$

3.5 BIC Objective

$$\text{BIC}(T) = \underbrace{T \cdot |\mathcal{C}| \cdot \ln V}_{\text{template cost}} + \underbrace{D(T) \cdot \ln(H \cdot |\mathcal{C}| \cdot V)}_{\text{deviation cost}}$$

Template cost $T \cdot |\mathcal{C}| \cdot \ln V$: encoding the template requires specifying $T \times |\mathcal{C}|$ values, each chosen from V possibilities. Grows linearly in T — longer periods

are penalized.

Deviation cost $D(T) \cdot \ln(H \cdot |C| \cdot V)$: each deviation is an anomaly encoded separately. The log factor is the cost of specifying which cell, which week, and which value it deviated to.

Optimal period:

$$T^* = \arg \min_{T \in [1, \min(13, H)]} \text{BIC}(T)$$

BIC PERIOD DETECTION · SITE-001 · H=13 · |C|=28 · V=20

Constants

$\ln(V)$	$= \ln(20)$	$= 3.00$
$\ln(H \cdot C \cdot V)$	$= \ln(13 \times 28 \times 20)$	$= \ln(7280) = 8.89$
Template cost per T-unit	$= C \times \ln(V) = 28 \times 3.00 = 84.0$	
Deviation cost per deviation	$= 8.89$	

Scoring each candidate period

T	Template cost	D(T)	Deviation cost	BIC(T)
1	$1 \times 84 = 84$	72	$72 \times 8.89 = 640$	724.0
2	$2 \times 84 = 168$	4	$4 \times 8.89 = 36$	204.0 ★
3	$3 \times 84 = 252$	18	$18 \times 8.89 = 160$	412.0
4	$4 \times 84 = 336$	14	$14 \times 8.89 = 124$	460.0

T* = 2 wins by a margin of 208 BIC points over T=3

WHERE DO THE 72 DEVIATIONS UNDER T=1 COME FROM?

Cell C08 (OR-102 Wed alternating): 6 deviations
→ 6 Patel weeks counted as deviations from Thompson modal

Cell C04 (OR-101 Thu, Garcia): 1 deviation
→ W01 Chen cover

Cell C05 (OR-101 Fri, Garcia): 1 deviation
→ W05 Wright cover

(64 more from other alternating cells at other sites)

Total under T=1: 72 deviations

WHY ONLY 4 DEVIATIONS UNDER T=2?

All alternating cells (C08 etc.) perfectly explained by 2 phases.
Only genuine one-off exceptions remain:

The offset k shifts the assignment: $\phi = (w + k) \bmod T^*$. We test all T^* candidate offsets and pick the one minimizing total deviations:

$$k^* = \arg \min_{k \in 0, 1, \dots, T^*-1} \sum_{c \in \mathcal{C}} \sum_{\phi=0}^{T^*-1} d_{c,\phi}^{(T^*,k)}$$

PHASE OFFSET DETECTION · SITE-001 · $T^*=2$ · Testing $k=0$ and $k=1$

Cell C08 (0R-102 Wednesday) history:

W00:Thompson W01:Patel W02:Thompson W03:Patel W04:Thompson
W05:Patel W06:Thompson W07:Patel W08:Thompson W09:Patel
W10:Thompson W11:Patel W12:Thompson

OFFSET $k=0 \rightarrow \text{Phase}(w) = w \bmod 2$

Phase 0 = {W00,W02,W04,W06,W08,W10,W12} $\rightarrow$ all Thompson $\rightarrow$ modal=Thompson
Phase 1 = {W01,W03,W05,W07,W09,W11} $\rightarrow$ all Patel $\rightarrow$ modal=Patel
Template: Phase 0 = Thompson | Phase 1 = Patel
Deviations: 0 ✓

OFFSET $k=1 \rightarrow \text{Phase}(w) = (w+1) \bmod 2$

Phase 0 = {W01,W03,W05,W07,W09,W11} $\rightarrow$ all Patel $\rightarrow$ modal=Patel
Phase 1 = {W00,W02,W04,W06,W08,W10,W12} $\rightarrow$ all Thompson $\rightarrow$ modal=Thompson
Template: Phase 0 = Patel | Phase 1 = Thompson
Deviations: 0 ✓

C08 alone cannot discriminate. Both offsets give 0 deviations here.

SUMMING ACROSS ALL 28 CELLS

Offset $k=0$: $D_{\text{total}} = 4$ (same 4 genuine exceptions as before)
Offset $k=1$: $D_{\text{total}} = 4$ (exceptions fall in different phases but same total count – tie)

Result: $k^* = 0$ (default on tie – no mathematical preference)

FINAL PHASE LABELS FOR SITE-001

Week	W00	W01	W02	W03	W04	W05	W06	W07	W08	W09	W10	W11	W12
Phase	0	1	0	1	0	1	0	1	0	1	0	1	0

Interpretation: Even-numbered weeks = Phase 0
Odd-numbered weeks = Phase 1

4.2 Design Note on Ties

When $k = 0$ and $k = 1$ produce equal deviation counts (common for $T^* = 2$ due to the symmetric alternating structure), we default to $k = 0$. The downstream impact is zero — Layer 1 learns phase-specific features regardless of which phase is labeled 0 vs 1. What matters is that the labeling is consistent within a run.

5. Canonical-Time Clustering (2-Hour Margin)

5.1 The Problem

Raw EHR block records contain minor time variation across weeks. A block nominally at 07:00–15:00 may appear as 07:05–15:05 in one week, 06:58–14:58 in another. Causes:

- Anesthesia setup time variation
- Room turnover from the previous case running long
- EHR system clocks occasionally out of sync
- Soft start policies allowing 5–15 minute flexibility

Without clustering, we get dozens of near-identical cells instead of one dense cell per logical slot. This inflates $|\mathcal{C}|$, inflates the BIC template cost, and can split a single logical block series into multiple fragments.

5.2 Algorithm

For each (site, room, day_of_week) group:

1. Collect all observed (start, end) pairs across all historical weeks
2. Cluster pairs where $|\text{start}_i - \text{start}_j| \leq 2\text{h}$ and $|\text{end}_i - \text{end}_j| \leq 2\text{h}$
3. For each cluster, select the **canonical time** as the most frequently observed (start, end) pair
4. Snap all other pairs in the cluster to the canonical time

CANONICAL-TIME CLUSTERING · OR-101 · Monday · 13 weeks

RAW OBSERVED (start, end) PAIRS

Week	Start	End	Δ from canonical
W00	07:02	15:00	+2 min start
W01	07:00	15:00	canonical ←
W02	07:05	15:03	+5 min start / +3 min end
W03	07:00	15:00	canonical ←
W04	07:00	15:00	canonical ←
W05	07:00	14:55	-5 min end
W06	07:00	15:00	canonical ←
W07	07:00	15:00	canonical ←
W08	06:58	14:58	-2 min start / -2 min end
W09	07:00	15:00	canonical ←
W10	07:00	15:00	canonical ←
W11	07:03	15:01	+3 min start / +1 min end
W12	07:00	15:00	canonical ←

CLUSTERING RESULT

Max pairwise Δ start = 7 min < 2 hours → one cluster
 Max pairwise Δ end = 8 min < 2 hours → one cluster

Frequency count:

(07:00, 15:00) → 8 occurrences ← CANONICAL
 (07:02, 15:00) → 1 occurrence
 (07:05, 15:03) → 1 occurrence
 (06:58, 14:58) → 1 occurrence
 (07:00, 14:55) → 1 occurrence
 (07:03, 15:01) → 1 occurrence

AFTER CLUSTERING

All 13 weeks snapped to (07:00, 15:00)
 Cell C01 = (OR-101 · Monday · 07:00–15:00) → 1 dense cell

WITHOUT CLUSTERING: 6 distinct "cells" for 1 logical slot
 Impact on BIC: |C| inflates 28 → 34 → template cost +18%
 Impact on T*: More noise in deviation count; may shift T*

5.3 Accuracy Improvement: Adaptive Threshold

The fixed 2-hour margin works well for 8-hour OR blocks but may over-merge genuinely distinct short slots. A more accurate threshold:

$$\text{threshold} = \min(2\text{h}, 0.25 \times \text{median_block_duration_of_room})$$

For a room with 4-hour median blocks: $\text{threshold} = \min(2\text{h}, 1\text{h}) = 1\text{h}$. For 8-hour median blocks: unchanged at 2h. This is a planned enhancement for sites with short-block scheduling.

6. Dominant Provider Assignment

6.1 The Modal Assignment

Once cells are canonical and phases are determined, each (cell, phase) pair gets a **dominant provider**:

$$m_{c,\phi}^{(T)} = \arg \max_{v \in \mathcal{V}} |\{w \in W_{T^*,\phi} : X[c][w] = v\}|$$

Ties broken by most recently observed value. The block template is the collection of all (cell, phase, dominant_provider) triples:

$$\mathcal{T} = \{(c, \phi, m_{c,\phi}^{(T^*)}) \mid c \in \mathcal{C}, \phi \in 0, \dots, T^* - 1\}$$

RECONSTRUCTED TEMPLATE · SITE-001 · T*=2 · k*=0 · H=13 weeks						
Block ID	Cell	Phase	Dominant Provider	Coverage		
BLK-001	OR-101 Mon 07:00	0	Garcia	7/7	100%	
BLK-002	OR-101 Mon 07:00	1	Garcia	6/6	100%	
BLK-003	OR-101 Tue 07:00	0	Chen	7/7	100%	
BLK-004	OR-101 Tue 07:00	1	Chen	6/6	100%	
BLK-005	OR-101 Wed 07:00	0	Garcia	7/7	100%	
BLK-006	OR-101 Wed 07:00	1	Garcia	6/6	100%	
BLK-007	OR-101 Thu 07:00	0	Garcia	7/7	100%	
BLK-008	OR-101 Thu 07:00	1	Garcia	5/6	83%	△
	└ 1 exception: Chen covered W01					
BLK-009	OR-101 Fri 07:00	0	Garcia	7/7	100%	
BLK-010	OR-101 Fri 07:00	1	Garcia	5/6	83%	△
	└ 1 exception: Wright covered W05					
BLK-011	OR-102 Mon 07:00	0	Thompson	7/7	100%	
BLK-012	OR-102 Mon 07:00	1	Thompson	6/6	100%	
BLK-013	OR-102 Tue 07:00	0	Kim	7/7	100%	

BLK-014	OR-102 Tue 07:00	1	Kim	6/6	100%
BLK-015	OR-102 Wed 07:00	0	Thompson	7/7	100%
BLK-016	OR-102 Wed 07:00	1	Patel	6/6	100%
└ KEY: clean rotation, Thompson Ph0 / Patel Ph1					
BLK-017	OR-102 Wed 07:00	0	Kim	6/7	86% ▲
└ 1 exception: Open W09 (scheduled maintenance)					
... (11 more blocks – unchanged stable assignments)					
BLK-028	OR-104 Mon 07:00	0	Open	7/7	100%
TEMPLATE SUMMARY					
Total template blocks : 28 (14 unique cells × 2 phases)					
100% coverage (no exceptions ever) : 22 blocks					
83–99% coverage (minor exceptions) : 4 blocks ← these carry signal					
Below 80% (genuinely contested) : 2 blocks ← flagged for review					

6.2 Open and Closed Blocks

"Open" and "Closed" are valid dominant providers. A cell consistently "Open" gets `dominant_provider="Open"` and is included in the template as an open block. A cell consistently "Closed" is excluded from optimization — no provider can be assigned to it.

A cell alternating between "Open" and a specific provider gets two template entries: one phase with the provider dominant, one phase with "Open" dominant.

7. Exception Rate Computation

7.1 Definition

An exception is any historical week where the actual blockholder differs from the dominant provider for that cell and phase:

$$\text{exceptions} = \{(c, w, X[c][w]) \mid X[c][w] \neq m_{c,\phi(w)}^{(T^*)}\}$$

Exception rate per series (one template slot across all its weeks = one series):

$$\text{exception_rate}[c, \phi] = \frac{|\{w \in W_{T^*, \phi} : X[c][w] \neq m_{c,\phi}^{(T^*)}\}|}{|W_{T^*, \phi}|}$$

7.2 What It Signals

Exception rate captures **slot contestedness** — how reliably a given block belongs to its dominant provider. High exception rate means the dominant provider frequently gives up that block; multiple providers compete for that slot; demand forecasting carries more uncertainty.

Exception rate feeds Layer 1 as a new XGBoost feature (higher rate → wider predicted variance). It also informs the Layer 2 consistency_score objective — contested slots get lower scores, penalizing allocations that assign them to providers with low historical presence.

EXCEPTION RATE ANALYSIS · SITE-001 · All blocks

FULL EXCEPTION LOG (4 exceptions in 13 weeks across 28 cells)

#	Block	Date	Dominant	Actual	Type
1	BLK-008	2026-01-13	Garcia	Chen	Provider substitution
	(OR-101 Thu, Phase 1, W01)				
2	BLK-010	2026-02-10	Garcia	Wright	Provider substitution
	(OR-101 Fri, Phase 1, W05)				
3	BLK-017	2026-03-03	Kim	Open	Maintenance close
	(OR-102 Wed, Phase 0, W09)				
4	BLK-024	2026-03-17	Jones	Thompson	Provider substitution
	(OR-103 Fri, Phase 1, W11)				

EXCEPTION RATES PER SERIES

Block ID	Cell	Ph	Exceptions	Weeks	Rate	Signal
BLK-001	OR-101 Mon 07:00	0	0	7	0.00	stable
BLK-002	OR-101 Mon 07:00	1	0	6	0.00	stable
BLK-007	OR-101 Thu 07:00	0	0	7	0.00	stable
BLK-008	OR-101 Thu 07:00	1	1	6	0.17	▲ contested
BLK-009	OR-101 Fri 07:00	0	0	7	0.00	stable
BLK-010	OR-101 Fri 07:00	1	1	6	0.17	▲ contested
BLK-015	OR-102 Wed 07:00	0	0	7	0.00	stable
BLK-016	OR-102 Wed 07:00	1	0	6	0.00	stable
BLK-017	OR-102 Wed 07:00	0	1	7	0.14	▲ maint.
BLK-024	OR-103 Fri 07:00	1	1	6	0.17	▲ contested
(all others)				—	0.00	stable

HOW EXCEPTION RATE AFFECTS DOWNSTREAM PREDICTIONS

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Garcia · Monday (BLK-001, rate=0.00):
  XGBoost feature input: exception_rate = 0.00
  Signal to model:       "This slot is always Garcia's.
                        Demand is predictable from his history."
  Bayesian posterior:    σ_pd narrow (e.g., ~8 min)

Garcia · Friday Phase 1 (BLK-010, rate=0.17):
  XGBoost feature input: exception_rate = 0.17
  Signal to model:       "This slot is Garcia's 83% of the time.
                        17% someone else holds it."
  Bayesian posterior:    σ_pd wider (e.g., ~25 min)
  Layer 2 impact:        consistency_score[Garcia][BLK-010] = 0.83
                        consistency_score[Wright][BLK-010] = 0.17

```

SUMMARY STATISTICS

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Total exception rate (site-wide): 4 / (28×13 / 2) ≈ 0.011
Blocks with rate = 0.00           : 24 / 28 (86%) highly stable
Blocks with rate ∈ (0, 0.20)      : 4 / 28 (14%) mildly contested
Blocks with rate > 0.20           : 0 / 28 (0%) no severely contested

```

7.3 Accuracy Improvement: Recency-Weighted Exception Rate

The current exception rate gives equal weight to all historical weeks. A recency-weighted variant:

$$\text{exception_rate_weighted}[c, \phi] = \frac{\sum_{w \in W_{T^*, \phi}} \rho^{H-1-w} \cdot \mathbf{1}[X[c][w] \neq m_{c, \phi}^{(T^*)}]}{\sum_{w \in W_{T^*, \phi}} \rho^{H-1-w}}$$

where $\rho \in (0, 1)$ is a decay factor (e.g., $\rho = 0.85$ per week). Recent exceptions count more than old ones. If Garcia started giving up Friday blocks more frequently in the last 4 weeks, this formula detects the trend while the unweighted version averages it away. Planned for a future release.

8. Accuracy Discussion and Design Alternatives

8.1 Limitations of the Current BIC Formulation

Cells are assumed independent. The BIC objective sums deviation costs across cells independently. In reality, cells within the same room on the same day are correlated — if OR-101 Monday is covered by Chen, that affects who covers OR-101 Tuesday that same week. A joint model would be more accurate but combinatorially harder. For most hospitals, independence is a reasonable approximation because block scheduling is done per-slot.

Global alphabet size V . Using $V = 20$ globally understates complexity savings from per-cell sparsity. Most cells have an effective alphabet of 2–4 providers. Using per-cell V_c would produce lower absolute BIC scores but the same relative ranking. Empirically validated: the same T^* is found with either approach.

Single global period. OR-101 might run a 2-week rotation while OR-104 runs a 4-week rotation. The algorithm finds one global T^* that minimizes total deviations. Room-specific period detection is a planned future enhancement for sites with heterogeneous room scheduling.

8.2 Alternative Period Detection Methods

Autocorrelation: For each cell, compute the autocorrelation of the binary assignment sequence (1 = dominant provider, 0 = other). The period is the lag with the highest autocorrelation peak, averaged across cells. Faster than BIC but less principled — does not penalize complexity and can overfit to noise.

Fourier analysis: Compute the DFT of each cell's binary history. The dominant frequency corresponds to the period. Equivalent to autocorrelation but more interpretable. Assumes stationarity; sensitive to missing weeks.

MDL (Minimum Description Length): A more principled information-theoretic alternative. MDL and BIC are asymptotically equivalent; MDL has better finite-sample properties. The BIC approximation is sufficient for $H \geq 10$ weeks.

Why BIC is the right choice: Computationally cheap (exhaustive search over at most 13 periods), interpretable, robust to noise, and empirically validated at

Geisinger ($T^* = 4$) and Atlanticare ($T^* = 2$). The alternatives offer marginal accuracy improvement at significant implementation complexity.

8.3 Validation Protocol

VALIDATION PROTOCOL FOR PERIOD DETECTION

Step 1 Obtain ground truth T_{true} from scheduling team
(ask OR committee or retrieve from block scheduling system)

Step 2 Run BIC on $H=13$ weeks.
Record T^* , $\text{BIC}(T^*)$, and $\text{BIC}(T_{\text{true}})$.

Step 3 Compute relative BIC gap:
 $\text{gap} = (\text{BIC}(T^*) - \text{BIC}(T_{\text{true}})) / \text{BIC}(T_{\text{true}})$

$T^* = T_{\text{true}} \rightarrow \text{gap} = 0.00 \quad \checkmark \quad \text{perfect}$

$T^* \neq T_{\text{true}} \rightarrow \text{gap} > 0 \quad \times \quad \text{investigate}$

Step 4 If $\text{gap} > 0.05$ (5%), investigate:
└ Is H large enough? Try extending to 26 weeks.
└ Structural break? Check rolling deviation rate.
└ True period > 13 weeks? Pass `template_weeks` explicitly.

Step 5 If T_{true} is known with confidence:
pass `template_weeks =  $T_{\text{true}}$` to override BIC for this site.

Step 6 Log result in `calibration.json` for cross-quarter tracking.
If T^* drifts between quarters, surface a warning.

SITE-001 VALIDATION RESULT (actual)

$T_{\text{true}} = 2$ (confirmed by scheduling team)

$T^* = 2$ (BIC detection result)

$\text{gap} = 0.00 \quad \checkmark$

$\text{BIC}(2) = 204$ vs $\text{BIC}(1)=724$, $\text{BIC}(3)=412$, $\text{BIC}(4)=460$

8.4 Handling Contested Dominance

A cell where the top two providers are close in frequency (e.g., Garcia holds 4/7 phase-0 weeks, Chen holds 3/7) is **ambiguously dominant**. Current behavior:

Garcia wins by plurality.

Better behavior: flag cells with dominance ratio < 0.60 (dominant appears fewer than 60% of the time) as contested and surface them in the UI for manual review. These cells should also receive a higher `exception_rate` feature value — they are structurally unstable, not just occasionally deviated.

Planned: `dominance_ratio` field in the `block_template` output, with a UI warning when ratio < 0.60 .

9. Outputs and Downstream Contracts

9.1 Full Output Specification

Output	Type	Downstream Consumer	Purpose
<code>T*</code>	int	Layer 1, Layer 2	Phase-alignment reference period
<code>phase_offset</code>	int	Layer 1, Layer 2	Canonical phase labeling
<code>phase_labels</code>	Dict<week → int>	Layer 1	Tags each historical week with rotation phase
<code>block_template</code>	List<Block>	Layer 2 warm-start	Dominant provider per cycling slot
<code>exceptions_list</code>	List<(block_id, date, provider_id)>	Layer 2 consistency scores	Full deviation record
<code>exception_rate_per_series</code>	Dict<block_id → float>	Layer 1 XGBoost feature	Per-series contestedness signal
<code>utilization_per_provider</code>	Dict<provider_id → float>	App UI	Historical performance
<code>bic_scores</code>	Dict<T → float>	App UI	Top-N period scores for override

Output	Type	Downstream Consumer	Purpose
warnings	List<string>	App UI	Data quality issues

9.2 Downstream Contract with Layer 1

Phase labels feed Stage 0 (Phase-Aligned History Synthesis). Each historical observation is tagged with its rotation phase. Trailing window features are computed within-phase: the 4-week trailing mean for a phase-0 week looks back at the 4 most recent phase-0 weeks only, not the 4 most recent calendar weeks.

Exception rate feeds Stage 2 (Feature Engineering) as a numeric XGBoost input per block_id. Higher value → XGBoost knows this slot is contested → predicted variance is higher → Bayesian model produces wider σ_{pd} .

9.3 Downstream Contract with Layer 2

Warm-start matrix: $W[p][b] = 1$ if provider p is the dominant provider for block b in the template. The MIP solver initializes from this warm-start, dramatically reducing solve time because the warm-start is usually near-optimal.

Consistency scores: The exceptions list identifies which non-dominant providers have historically held each series. These providers receive lower (but non-zero) consistency scores if assigned to those blocks.

DOWNSTREAM CONTRACT WALKTHROUGH · Garcia's Friday Block (BLK-010)	
PRE-LAYER OUTPUT FOR BLK-010	
block_id	: BLK-010
cell	: OR-101 · Friday · 07:00–15:00
phase	: 1 (odd weeks)
dominant_provider	: Garcia (5/6 = 83%)
room_type	: Cardiac
exception_rate	: 0.17 (Wright covered W05)
exceptions_list	: [{week:5, date:2026-02-10, actual:Wright}]
WHAT LAYER 1 RECEIVES	
Feature fed to XGBoost for Garcia × Friday predictions:	

```

exception_rate_series = 0.17

Effect on model:
  XGBoost: "Garcia's Friday slot is contested 17% of the time.
           Historical demand in this slot is less reliable."
  Bayesian:  $\sigma_{pd}$ [Garcia, Fri] wider than  $\sigma_{pd}$ [Garcia, Mon]
           Monday = 0.00 exception rate  $\rightarrow \sigma_{pd} \approx 8$  min
           Friday = 0.17 exception rate  $\rightarrow \sigma_{pd} \approx 25$  min
  Scenarios: P10-P90 band for Garcia×Friday is ~3× wider than Monday

WHAT LAYER 2 RECEIVES


---


Warm-start:
  W[Garcia][BLK-010] = 1   (Garcia is the dominant holder)

Consistency scores for BLK-010 series:
  Garcia  $\rightarrow$  held 5/6 phase-1 Friday instances  $\rightarrow$  score = 0.83
  Wright  $\rightarrow$  held 1/6 phase-1 Friday instances  $\rightarrow$  score = 0.17
  All others  $\rightarrow$  score = 0.00 (never held this slot)

Optimizer decision:
  If the goal is to remove Garcia's blocks (target util 70%),
  and the optimizer considers reassigning BLK-010 to Wright:
  └ 05 consistency: Wright scores 0.17 vs Garcia 0.83
  └ Room compatibility: Wright is Cardiac ✓
  └ C5 site exclusivity: Wright is SITE-001 only ✓
  └ Result: Wright can receive BLK-010 but at consistency penalty

```

9.4 Full API Response

```

{
  "T_star": 2,
  "phase_offset": 0,
  "phase_labels": {
    "0": 0, "1": 1, "2": 0, "3": 1, "4": 0,
    "5": 1, "6": 0, "7": 1, "8": 0, "9": 1,
    "10": 0, "11": 1, "12": 0
  },
  "bic_scores": {
    "T1": 724.0, "T2": 204.0, "T3": 412.0, "T4": 460.0
  },

```

```

"block_template": [
  {
    "block_id": "BLK-010",
    "site": "SITE-001",
    "occurrence": {
      "week_num": 1,
      "day": "Friday",
      "start": "07:00",
      "end": "15:00"
    },
    "current_blockholder": {"provider_id": "PRV-003"},
    "room": {"type": "Cardiac"},
    "dominance_ratio": 0.83
  }
],
"exception_rate_per_series": {
  "BLK-001": 0.00, "BLK-002": 0.00,
  "BLK-008": 0.17, "BLK-010": 0.17,
  "BLK-017": 0.14, "BLK-024": 0.17
},
"exceptions_list": [
  {
    "block_id": "BLK-008",
    "week": 1,
    "date": "2026-01-13",
    "dominant_provider": "PRV-003",
    "actual_provider": "PRV-001"
  },
  {
    "block_id": "BLK-010",
    "week": 5,
    "date": "2026-02-10",
    "dominant_provider": "PRV-003",
    "actual_provider": "PRV-006"
  },
  {

```

```

    "block_id": "BLK-017",
    "week": 9,
    "date": "2026-03-03",
    "dominant_provider": "PRV-007",
    "actual_provider": "OPEN"
  },
  {
    "block_id": "BLK-024",
    "week": 11,
    "date": "2026-03-17",
    "dominant_provider": "PRV-005",
    "actual_provider": "PRV-015"
  }
],
"providers": [
  {
    "provider_id": "PRV-003",
    "name": "Dr. James Garcia",
    "service_lines": ["Cardiac"],
    "exclusive_sites": ["SITE-001"],
    "total_in_block_case_minutes": 198,
    "total_out_of_block_case_time_minutes": 0,
    "previous_blocks_count": 28,
    "current_utilization": 0.026,
    "total_block_case_minutes": 162,
    "total_turnover_minutes": 36,
    "total_early_release_minutes": 1840
  }
],
"warnings": []
}

```

Next section: Layer 1 — Stochastic Demand Forecasting →

Document Owner: Zernitsky Itamar **Last Updated:** May 2026